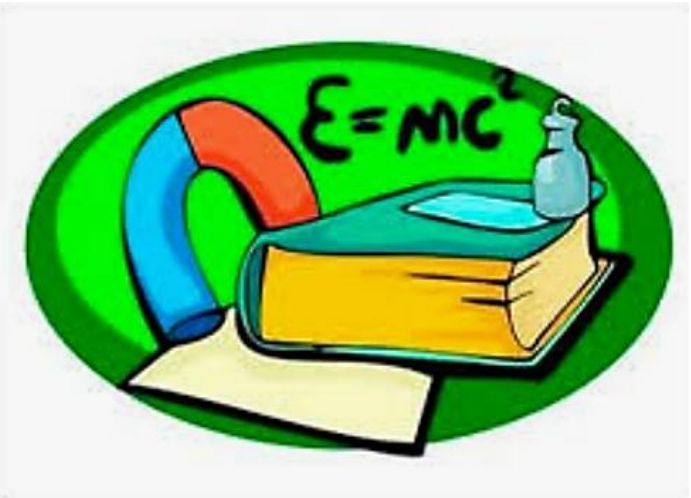
# Fizika

9-sinf

## Mavzu: **Masalalar yechish.**



$$A_{foy.} = Q_1 - Q_2$$

$$\eta = \frac{A_{foy.}}{Q_1} \cdot 100\ \% = \frac{Q_1 - Q_2}{Q_1} \cdot 100\ \%$$

$$\eta = \frac{T_1 - T_2}{T_1} \cdot 100\ \%$$

# Test

Karno sikli qanday jarayonlardan tashkil topgan?

A) Izotermik, izobarik, izoxorik va adiabatik.

B) 2 ta izotermik va 2 ta adiabatik.

C) 2 ta izotermik va 2 ta izobarik.

D) 2 ta adiabatik va 2 ta izoxorik.

**To‘g‘I javob: B) 2 ta izotermik va 2 ta adiabatik.**

# 18-mashq 2-masala

Isitkichning temperaturasi 477 ℃, sovitkichniki 27 ℃ bo‘lgan issiqlik mashinasining maksimal FIK ni hisoblang.

**Berilgan:**

$t_1 = 477°\text{C}$

$t_2 = 27°\text{C}$

**Topish kerak:** $\eta - ?$

**Formula:**

$$T = t + 273$$

$$\eta = \frac{T_1 - T_2}{T_1} \cdot 100\ \%$$

**Yechish:**

$$T_1 = 477°\text{C} + 273 = 750\ \text{K}$$

$$T_2 = 27°\text{C} + 273 = 300\ \text{K}$$

$$\eta = \frac{750\ \text{K} - 300\ \text{K}}{750\ \text{K}} \cdot 100\ \% = 60\ \%$$

**Javob:** $\eta = 60\ \%$

# 18-mashq 4-masala

Agar isitkichning temperaturasi $127℃$, sovitkichniki $7℃$ bo‘lgan ideal issiqlik mashinasi bir siklda isitkichdan $1300\,J$ issiqlik olsa, bajariladigan foydali ish nimaga teng?

**Berilgan:**

$t_1 = 127℃$

$t_2 = 7℃$

$Q = 1300\ \text{J}$

**Topish kerak:**

$A_f - ?$

**Formula:**

$$T = t + 273$$

$$\eta = \frac{T_1 - T_2}{T_1} \cdot 100\ \% = \frac{A_f}{Q} \cdot 100\ \%$$

$$A_f = \frac{Q \cdot (T_1 - T_2)}{T_1}$$

**Yechish:**

$$T_1 = 127℃ + 273 = 400\ \text{K}$$

$$T_2 = 7℃ + 273 = 280\ \text{K}$$

$$A_f = \frac{1300\ J \cdot (400\ K - 280\ K)}{400\ K} = 390\ \text{J}$$

**Javob: $A_f = 390\ \text{J}$**

# 18-mashq 6-masala

Ideal issiqlik dvigatelida sovitkichning temperaturasi 62℃, issiqlik dvigatelining FIK 50 % bo‘lsa, isitkich bilan sovitkich temperaturalari orasidagi farq qanday?

## Berilgan:

$t_2 = 62°C$

$\eta = 50\ \%$

## Topish kerak:

$\Delta T - ?$

## Formula:

$$T = t + 273 \qquad \Delta T = T_1 - T_2$$

$$\eta = \frac{T_1 - T_2}{T_1} \cdot 100\% \Rightarrow \eta T_1 = (T_1 - T_2) \cdot 100\ \%$$

$$\eta T_1 = T_1 \cdot 100\ \% - T_2 \cdot 100\ \%$$

$$T_2 \cdot 100\ \% = T_1 \cdot 100\ \% - \eta T_1$$

$$T_1 = \frac{T_2 \cdot 100\ \%}{100\ \% - \eta} \Rightarrow \Delta T = \frac{T_2 \cdot 100\ \%}{100\ \% - \eta} - T_2$$

**Yechish:** $T_2 = 62℃ + 273 = 335\ \text{K}$

$$\Delta T = \frac{335\ \text{K} \cdot 100\ \%}{100\ \% - 50\ \%} - 335\ \text{K} = 335\ \text{K}$$

**Javob:** $\mathbf{\Delta T = 335\ K}$

# 5-test 91-bet

Ideal issiqlik mashinasining foydali ish koeffitsiyenti 75 % bo‘lishi uchun isitkichning temperaturasi sovitkichning temperaturasidan necha marta katta bo‘lishi kerak?

A) 4; B) 3; C) 5; D) 2.

## Berilgan:

$\eta = 75\%$

$\frac{T_1}{T_2} - ?$

## Formulasi:

$\eta = \frac{T_1 - T_2}{T_1} \cdot 100\%$

$\eta = 100\% - \frac{T_2}{T_1} \cdot 100\%$

$\frac{T_2}{T_1} \cdot 100\% = 100\% - \eta$

$\frac{T_1}{T_2} = \frac{100\%}{100\% - \eta}$

## Yechish:

$\frac{T_1}{T_2} = \frac{100\%}{100\% - 75\%} = 4$

**Javob: A) 4**

# Mustaqil bajarish uchun topshiriqlar

18-mashqning  3, 5, 7, 8- masalalarini yechish. 87-bet.